LUC@BJTU Weihai Campus

Fundamentals of Communication Systems (CNSCC160) Tutorial 2

B

1. What is a signal in the context of communication systems?

a) A function of multiple dependent variables
b) A function of one or more independent variables containing information
c) A process with no temporal variation
d) A static electrical circuit

A

2. Which form represents a signal as a voltage source?

a) Thevenin's form
b) Norton's form
c) Kirchhoff's form
d) Coulomb's form

C

3. What type of signal is defined for every instant of time within a range?

a) Discrete-time signal
b) Pulse signal
c) Continuous-time signal
d) Sine wave signal

B

4. A discrete-time signal is represented as:

a) x(t) for continuous values of t
b) x[n], where n is a sequence of numbers
c) y(t) for finite values
d) z[n], for complex numbers only

D

5. Which of the following is NOT an example of an analogue signal?

a) Voice
b) Sine wave
c) Video
d) Serial binary code

C

6. A signal pattern that repeats over time is known as a:

a) Aperiodic signal
b) Discrete signal
c) Periodic signal
d) Modulated signal

B

7. What is the period (T) of a signal related to?

a) The speed of propagation
b) The signal frequency
c) The amplitude of the signal
d) The wavelength of the medium

C

8. The range of frequencies a signal contains is known as the:

a) Wavelength
b) Period
c) Spectrum
d) Phase

C

9. Which formula is used to obtain a signal's frequency spectrum?

a) Nyquist formula
b) Shannon's capacity
c) Fourier series
d) Euler's theorem

C

10. What does Signal-to-Noise Ratio (SNR) quantify?

a) The bandwidth of a channel
b) The amount of signal distortion
c) The ratio of signal power to noise power
d) The total power of a system

D

11. Which of the following is true about a noiseless channel?

a) It allows zero transmission errors
b) It is described by Shannon's capacity

c) It cannot transmit high bit rates
d) Its bit rate is determined by the Nyquist formula

B

12. Which of the following transmission media requires line-of-sight?

a) Optical fiber
b) Microwave
c) Radio waves
d) Copper wire

C

13. Which type of wave cannot penetrate walls?

a) Radio waves
b) Microwaves
c) Infrared waves
d) Visible light

B

14. What is the purpose of multiplexing in communication systems?

a) To compress data for easier storage
b) To increase the number of signals transmitted over a single channel
c) To enhance signal quality by amplifying the signal
d) To separate signals by increasing their frequency

D

15. What type of multiplexing uses time slots to transmit multiple signals?

a) Frequency-division multiplexing (FDM)
b) Code-division multiplexing (CDM)
c) Space-division multiplexing (SDM)
d) Time-division multiplexing (TDM)